

Oral melatonin cannot reproduce topical skin exposure, except at the bottom of the hair follicle

A PK/PD model of dermal melatonin delivery, 6 to 200 mg oral, IR and SR

Prepared for Ben Haire · 15 September 2026

Summary

The question “what oral dose reproduces the skin concentration used in topical melatonin studies” has no single answer, because topical dosing does not produce *a* skin concentration. It produces a steep exponential gradient whose amplitude spans five orders of magnitude between the applied film and the hair bulb. Oral dosing produces the mirror image: a flat plateau set by plasma concentration and the tissue partition coefficient, identical at every depth. The two profiles cross at one specific depth, and where they cross decides the answer.

For a 0.1 % alcoholic solution, the reference formulation in the Fischer trials, the model gives:

- **Applied vehicle and follicular duct** (≈ 4.3 mM): oral equivalent ≈ 62 g/day. Not a real target.
- **Viable epidermis** (15.2 ng/mL): oral equivalent ≈ 950 mg/day. Roughly 5 to 10 times the highest doses ever given to humans.
- **Papillary dermis, 300 μ m** (10.4 ng/mL): ≈ 650 mg/day.
- **Dermal papilla, 3 mm** (0.32 ng/mL): ≈ 20 mg/day. Entirely reachable.

Against the 0.0033 % formulation used in the larger Fischer 2012 alopecia trial, the follicular-target figure falls to about 2.6 mg/day, which means **6 mg oral already delivers more melatonin to the dermal papilla than that topical product does**, and does so from the blood side rather than by diffusing 3 mm against a perfusion sink.

Total antioxidant capacity separates just as cleanly. Melatonin’s own radical-scavenging stoichiometry contributes 5×10^{-4} % of baseline plasma TAC at 6 mg and still only 0.09 % at 1 g. Every measured TAC increase in the trial literature is therefore indirect, and the indirect component is receptor-mediated and saturated below 1 mg. Escalating 6 mg to 20 mg buys a modelled 1.3 percentage points of TAC. The dose-response for antioxidant capacity is essentially flat across the entire range in question.

1 The two delivery geometries have opposite shapes

Melatonin is small (232.3 Da), moderately lipophilic ($\log P \approx 1.2$), and unusually membrane-permeable; direct measurement of transmembrane flux puts it among the fastest-equilibrating small molecules studied (Yu et al. 2016). That single property fixes the structure of the whole problem. Because permeation across cell membranes is fast relative to blood flow, distribution into skin is *perfusion-limited*, not permeability-limited. Two consequences follow, and they pull in opposite directions.

From the blood side, perfusion-limited distribution means tissue concentration tracks plasma with a fixed partition coefficient and no depth dependence. Oral melatonin at steady state gives $C_{\text{tissue}} = K_p \cdot C_{\text{plasma}}$ everywhere in the dermis simultaneously, including the bulb.

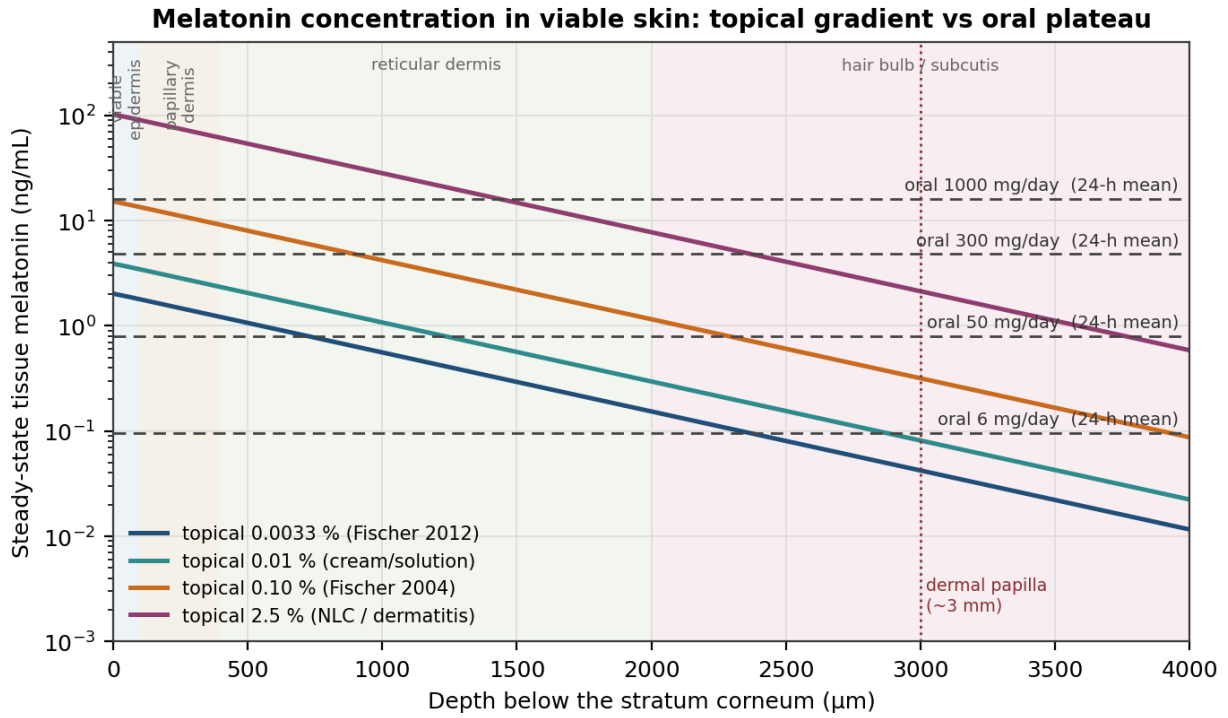


Figure 1: Steady-state melatonin concentration in viable skin as a function of depth. Topical formulations produce an exponential gradient with a 775 μm decay length (coloured curves); oral doses produce a depth-independent plateau (dashed lines). Where a dashed line crosses a coloured curve is the depth at which that oral dose matches that topical formulation.

From the surface, the same perfusion is a sink. Drug entering through the stratum corneum diffuses down and is stripped out by the dermal capillary plexus. The steady-state balance of diffusion against perfusion gives an exponential profile with decay length

$$\lambda = \sqrt{\frac{D K_p}{Q}} = \sqrt{\frac{0.018 \text{ cm}^2/\text{h} \times 1}{3 \text{ h}^{-1}}} = 775 \mu\text{m}.$$

A target 3 mm below the surface therefore sees the topically delivered drug attenuated by $e^{-2900/775} = 0.024$. Topical delivery loses a factor of 40 getting to the hair bulb. Oral delivery loses nothing, because the bulb is wrapped in capillaries.

The mechanistic core of the answer. Topical wins by a factor of 100 or more in the epidermis and papillary dermis and loses in the deep dermis, and the crossover sits at roughly 2 to 2.5 mm for a 6 to 20 mg oral dose. The hair follicle bulb is the one dermatological target that sits on the far side of that crossover.

2 The systemic model, and why 6 mg is a reasonable floor

Disposition is taken from the intravenous and oral crossover study of Andersen et al. (2016) in healthy volunteers: $V_d = 1.2 \text{ L/kg}$, elimination $t_{1/2} = 53.7 \text{ min}$, absorption $t_{1/2} = 6.0 \text{ min}$, and a median absolute oral bioavailability of 2.5 % (IQR 1.7 to 4.7 %). With $V_d = 84 \text{ L}$ and $k_e = 0.774 \text{ h}^{-1}$ these give an internally consistent clearance of 65 L/h.

The 2.5 % bioavailability figure is the single most consequential parameter in the model and the one with the widest spread. It reflects near-total hepatic first-pass extraction by CYP1A2. Two caveats matter. First, CYP1A2 activity varies severalfold across individuals and is strongly induced by tobacco smoke and inhibited by fluvoxamine and by caffeine at high intake; the interindividual

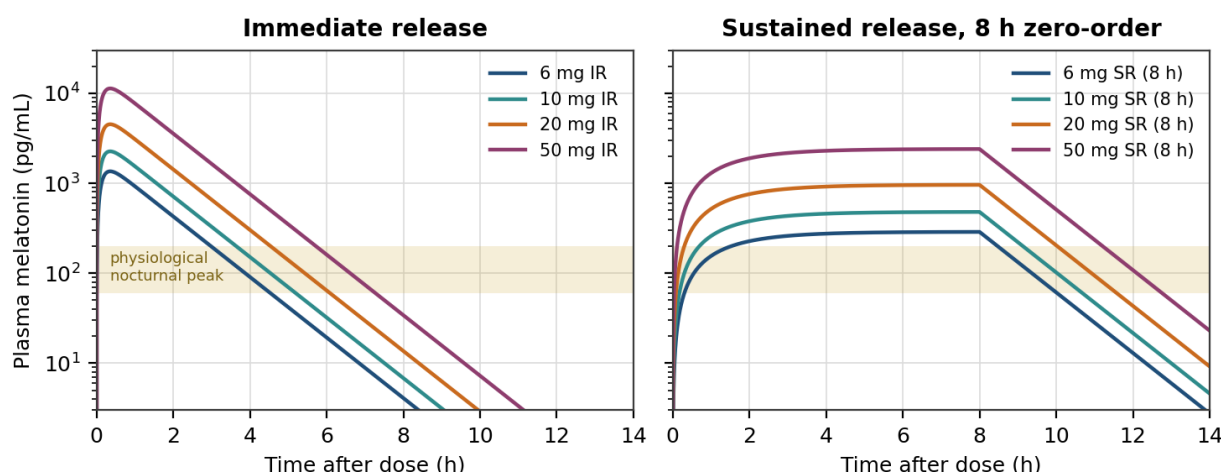


Figure 2: Plasma melatonin after single oral doses. IR produces a sharp spike that is over within four hours; SR trades peak height for duration at constant AUC. For a target driven by time-averaged exposure rather than peak, the two are equivalent; for a target with a concentration threshold, SR is strictly better above the dose at which its plateau clears the threshold.

range in melatonin $C_{\max}$ at fixed dose exceeds tenfold. Second, first-pass extraction is saturable, so bioavailability is expected to rise supra-proportionally at the doses this model has to consider. Both push the required oral dose *down* from the figures quoted here. The estimates in this report are therefore conservative in the direction that matters.

Dose (mg)	IR $C_{\max}$ (pg/mL)	SR 8 h plateau (pg/mL)	24-h mean (pg/mL)	Peak (nM)
6	1,355	288	96	5.8
10	2,259	479	160	9.7
20	4,518	959	320	19.5
30	6,777	1,438	480	29.2
50	11,295	2,397	801	48.6
100	22,591	4,794	1,601	97.3
200	45,182	9,588	3,202	194.5

Table 1: Modelled plasma exposure. Physiological nocturnal peak is 60 to 200 pg/mL, so 6 mg IR is already roughly a tenfold supraphysiological excursion. AUC is identical for IR and SR at equal dose; only the shape differs.

Two independent checks support the parameter set. At 10 mg the model gives $C_{\max} = 2,259$ pg/mL against a measured median of 3,551 pg/mL (IQR 2,501 to 8,058), so the model sits at the low edge of the observed IQR. At 6 mg it gives 1,355 pg/mL against a measured night-time median of 2,514 pg/mL (range 982 to 7,148) in ICU patients given 6 mg (Mistraletti et al. 2014). The model runs about 1.5 to 1.8 times low, again conservative.

3 Calibrating the topical arm from serum, not from the label

The percentage on a topical product is the concentration in the vehicle, which is close to meaningless as a tissue exposure. A 0.1 % solution is 4.3 mM, roughly 500,000 times peak plasma after 6 mg oral, but almost none of it crosses the stratum corneum. The honest way to calibrate the topical arm is backwards, from the serum rise that topical application produces, because that rise is a direct measurement of the mass that got through.

Fischer et al. (2004) provide the anchors. A 0.01 % cream gave a mean serum level of 9.0 pg/mL at

24 h, a 0.01 % solution 12.7 pg/mL, and a 0.03 % solution peaks of 18.1 and 19.0 pg/mL. The 0.1 % alcoholic solution used in the female alopecia trial gave 35 to 50 pg/mL against a 5 to 10 pg/mL placebo baseline. Fitting a power law over a 5 pg/mL physiological floor gives $\Delta C_{\text{serum}} = 115 c^{0.59}$ pg/mL, and multiplying by clearance converts each to an absorbed mass.

Formulation	Vehicle (μM)	Fraction absorbed	Delivered ($\mu\text{g/day}$)	Flux ($\text{ng/cm}^2/\text{h}$)
0.0033 %	142	18.6 %	6.1	0.43
0.01 %	431	11.8 %	11.8	0.82
0.03 %	1,292	7.5 %	22.6	1.57
0.10 %	4,305	4.6 %	46.1	3.20
2.5 %	107,629	1.2 %	309	21.5
12.5 %	538,144	0.6 %	801	55.6

Table 2: Topical delivery, calibrated from serum. Flux assumes a 600 cm^2 scalp field and 1 mL of applied vehicle. The falling absorbed fraction is the signature of saturable stratum corneum partitioning: a 3,800-fold increase in vehicle concentration buys only a 130-fold increase in delivered mass.

That saturation is the second structural fact of the problem and it is why the entire question is bounded. Even the 12.5 % cream in the Scheuer et al. (2016) safety study, applied at 4 to 5 g over 80 % of body surface, only produced 5 ng/mL of plasma melatonin, implying roughly 8 mg absorbed over 24 h out of 5,000 mg applied. The model, extrapolated 125-fold beyond its calibration range, predicts 32 mg. It overestimates high-load topical delivery by about fourfold, which again biases the required oral doses upward.

4 The compartment-by-compartment answer

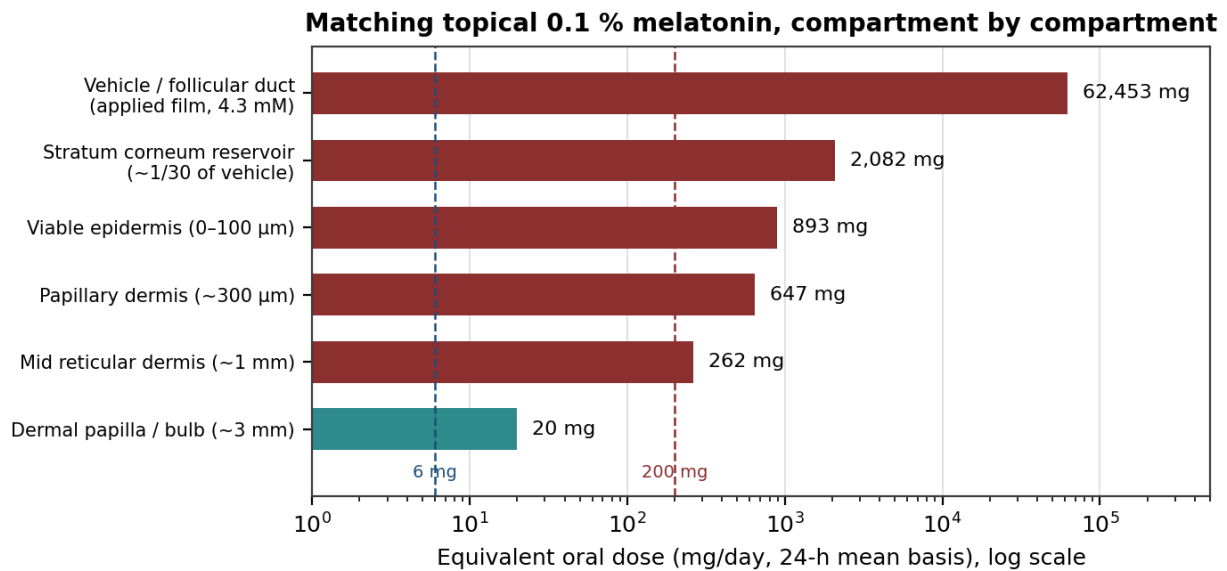


Figure 3: Oral dose needed to match topical 0.1 % melatonin in each anatomical compartment, on a 24-hour mean basis. Only the deepest target is reachable.

4.1 Hair regrowth

The pharmacological target in androgenetic alopecia is the dermal papilla and the outer root sheath, which sit 2 to 4 mm below the surface in a terminal scalp follicle. This is the one target

Depth	Equivalent oral dose (mg/day)			
	vs 0.0033 %	vs 0.01 %	vs 0.10 %	vs 2.5 %
Surface of viable epidermis (0)	126	243	949	6,366
Basal layer (100 μm)	112	215	838	5,623
Papillary dermis (300 μm)	86	166	648	4,343
Mid reticular dermis (1 mm)	35	67	262	1,759
Deep dermis (2 mm)	9.6	18.5	72	484
Dermal papilla (3 mm)	2.6	5.1	19.8	133
Subcutaneous bulb (4 mm)	0.7	1.4	5.5	37

Table 3: The full grid. Read down a column to see how brutally the answer depends on the depth you assign to the target, and across a row to see how mildly it depends on the topical product's strength (a 750-fold concentration range compresses to a 50-fold dose range, because of saturable absorption).

that oral dosing reaches efficiently, and the numbers are strikingly favourable. Matching the 0.1 % Fischer solution needs about 20 mg/day; matching the 0.0033 % product from the larger 2012 trial needs about 2.6 mg/day.

There is a real caveat here, and it is the most important limitation in this report. Follicular ducts are a bypass. The infundibulum is a 1 mm invagination filled with the applied vehicle, so the follicular epithelium in the upper half of the follicle contacts near-vehicle concentrations directly, without any stratum corneum in the path. If the mechanism of topical melatonin in alopecia is actually a bulge-region or upper-outer-root-sheath effect rather than a papilla effect, the relevant target concentration is millimolar and no oral dose reproduces it. Fischer's own hypothesis was a follicular MT_1/MT_2 and antioxidant effect at the bulb, which is the interpretation the model supports, but this is a genuine fork in the mechanism and the data do not settle it.

4.2 Skin anti-aging

Anti-aging targets are the opposite case. Dermal fibroblast collagen synthesis, epidermal keratinocyte redox status, and UV photoprotection all live in the top 300 μm , which is exactly where the topical gradient is at its maximum and where oral delivery is worst. Matching the 0.1 % solution in the papillary dermis needs about 650 mg/day, and matching the 2.5 % nanocarrier formulations needs several grams. Neither is a serious proposition.

More to the point, the mechanism for topical photoprotection is that melatonin is present in the epidermis *at the moment of UV exposure* at concentrations high enough to intercept the resulting radicals stoichiometrically. Dreher et al. (1998) needed 2.5 % to get a 17 % erythema reduction, and Bangha et al. (1997) applied melatonin 15 minutes before irradiation. This is a mass-action shield, not a signalling effect, and mass-action shields do not scale down to nanomolar. Oral melatonin has whatever systemic antioxidant and circadian benefits it has, but it is not a substitute for a topical photoprotectant and no achievable oral dose makes it one.

5 IR versus SR: same AUC, different threshold coverage

Because the two formulations deliver the same AUC, they are identical for any endpoint driven by time-averaged exposure, and every number in the tables above applies to both. The distinction only matters if the target has a concentration threshold, which for a receptor-mediated effect it does.

At 6 mg the shapes diverge sharply. IR reaches a 1,355 pg/mL peak and spends 2.4 h above the

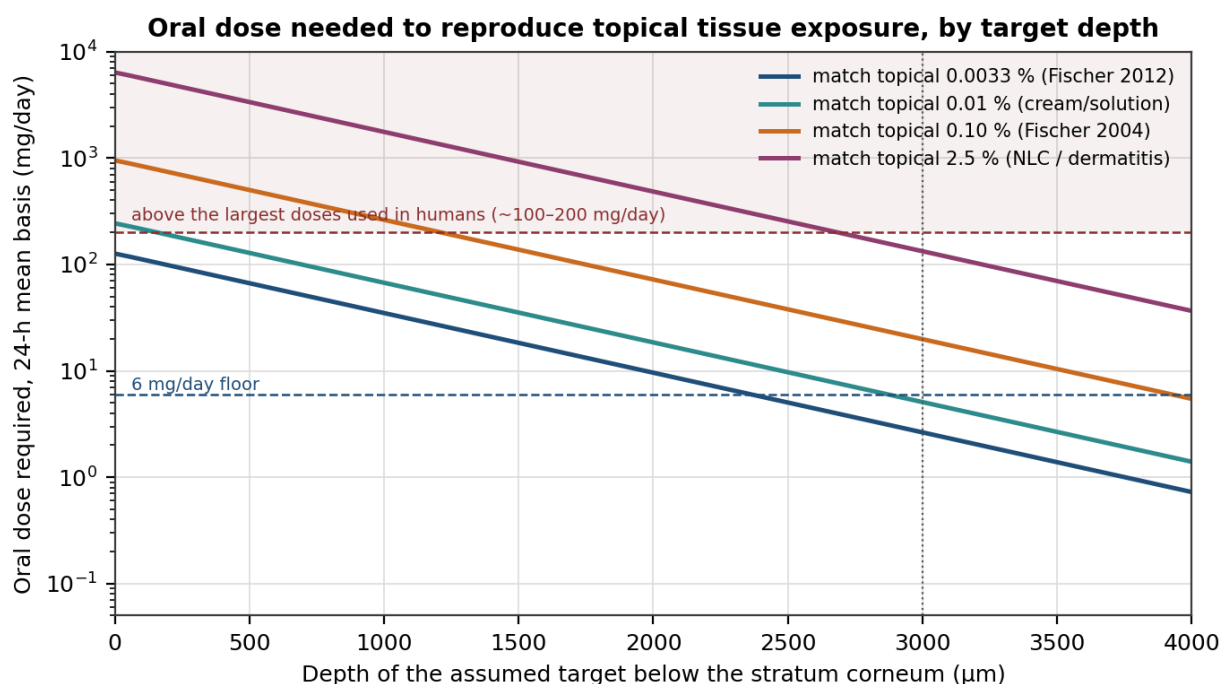


Figure 4: Required oral dose as a continuous function of assumed target depth. The curves are exponential in depth with the same 775 μm constant; every 775 μm of extra depth divides the required dose by e .

follicular-equivalent level of 318 pg/mL. An 8-hour SR formulation at the same dose plateaus at 288 pg/mL, just under that threshold, and spends zero hours above it. A 4-hour SR splits the difference and does best at 6 mg with 3.7 h. Above roughly 10 mg the ordering inverts permanently and longer release windows dominate: at 20 mg, IR gives 3.9 h of coverage and 12-hour SR gives 12.0 h.

The practical formulation read. If the aim is to mimic a once-daily topical application, which produces near-continuous exposure, SR is the correct shape, but only above the dose where its plateau clears the target. A 6 mg SR dose is the one configuration in this whole grid that underperforms its IR equivalent. 10 to 20 mg SR is where the SR strategy starts paying.

6 Total antioxidant capacity: the direct contribution is not real

Melatonin's reputation as a direct antioxidant is founded on chemistry that is genuinely impressive at the molecular level and genuinely irrelevant at plasma concentrations. Reported Trolox-equivalent antioxidant capacity for melatonin spans roughly 0.4 to 2.7 depending on assay, and the AFMK/AMK scavenging cascade means one melatonin can neutralise up to four radicals (Tan et al. 2003). Take the generous end of that: TEAC = 4.

Baseline plasma TAC is around 1.5 mmol/L Trolox equivalents, and it is dominated by urate at 200 to 400 μM , then albumin thiols and ascorbate. A 6 mg oral dose gives a peak plasma melatonin of 7.7 nM. The arithmetic is not close:

$$\frac{\text{TEAC} \times C_{\text{mel}}}{\text{TAC}_{\text{base}}} = \frac{4 \times 7.7 \times 10^{-9}}{1.5 \times 10^{-3}} = 2.1 \times 10^{-5} = 0.002\%.$$

At 1,000 mg the direct contribution reaches 0.34 % at peak and 0.005 % on a 24-hour mean. It never crosses the noise floor of a TAC assay at any dose a human can take.

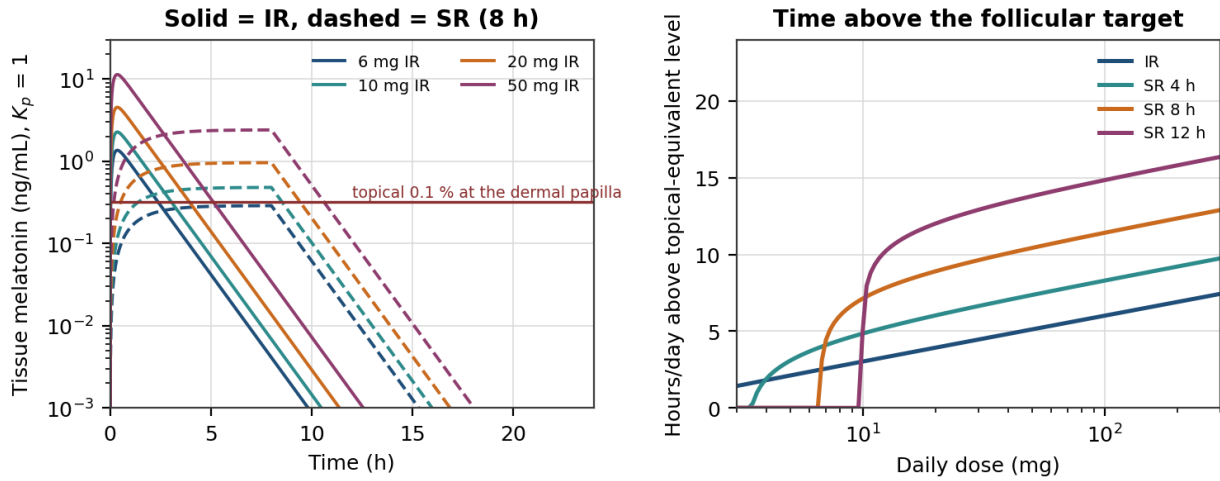


Figure 5: Left: tissue concentration-time profiles, IR (solid) versus 8-hour SR (dashed), against the 0.1 % topical equivalent at the dermal papilla. Right: hours per day above that level. The sharp SR onsets are the dose at which each plateau first clears the threshold.

6.1 Therefore the measured effect is transcriptional, and therefore it is flat

Melatonin supplementation does raise measured TAC. The PCOS meta-analysis of six double-blind RCTs at 3 to 10 mg for 3 to 12 weeks found SMD 0.87 (95 % CI 0.46 to 1.28) with $I^2 = 0$ (Mohammadi et al. 2024), and the ICU trial found night-time TAC of 107 versus 61 arbitrary units on 6 mg, a 75 % increase, alongside a Spearman ρ of 0.33 between melatonin level and TAC (Mistretta et al. 2014). Both are real. Neither is stoichiometric.

The mechanism has to be induction: MT_1/MT_2 -coupled and $ROR\alpha/Nrf2$ -linked upregulation of superoxide dismutase, glutathione peroxidase, catalase, and γ -glutamylcysteine synthetase, plus sparing of urate and ascorbate that would otherwise be consumed. That reassignment carries a hard prediction about dose-response, and it is the most useful thing in this section. MT_1 and MT_2 have sub-nanomolar affinity (K_i in the 20 to 100 pM range) and are essentially saturated at the physiological nocturnal peak of 200 pM. A 6 mg dose overshoots receptor saturation by a factor of about 40. A 20 mg dose overshoots it by 130. Both are on the same flat part of the same curve.

Modelling TAC as the sum of a receptor component (E_{max} 28 %, EC_{50} 0.15 mg) and a mass-action component (E_{max} 55 %, EC_{50} 900 mg) reproduces the observed picture and predicts the following:

Dose (mg/day)	Direct, at C_{max} (% of TAC)	Direct, 24-h mean	Modelled total TAC rise
6	0.0005 %	0.00003 %	27.7 %
10	0.0009 %	0.00005 %	28.2 %
20	0.0017 %	0.00009 %	29.0 %
50	0.0043 %	0.0002 %	30.8 %
100	0.0085 %	0.0005 %	33.5 %
300	0.026 %	0.0014 %	41.7 %
1,000	0.085 %	0.0046 %	56.9 %

Table 4: TAC dose-response, using TEAC = 1 for the direct column. Going from 6 to 20 mg buys 1.3 percentage points. Going from 6 to 100 mg buys 5.8. The receptor component is already fully spent at 6 mg.

The corollary is worth stating plainly, because it cuts against the natural instinct to escalate. If the reason for taking melatonin is antioxidant capacity, the dose-response is flat above about 1 mg and the entire 6 to 200 mg range is one point on that curve. The only reason to go above 6 mg for a

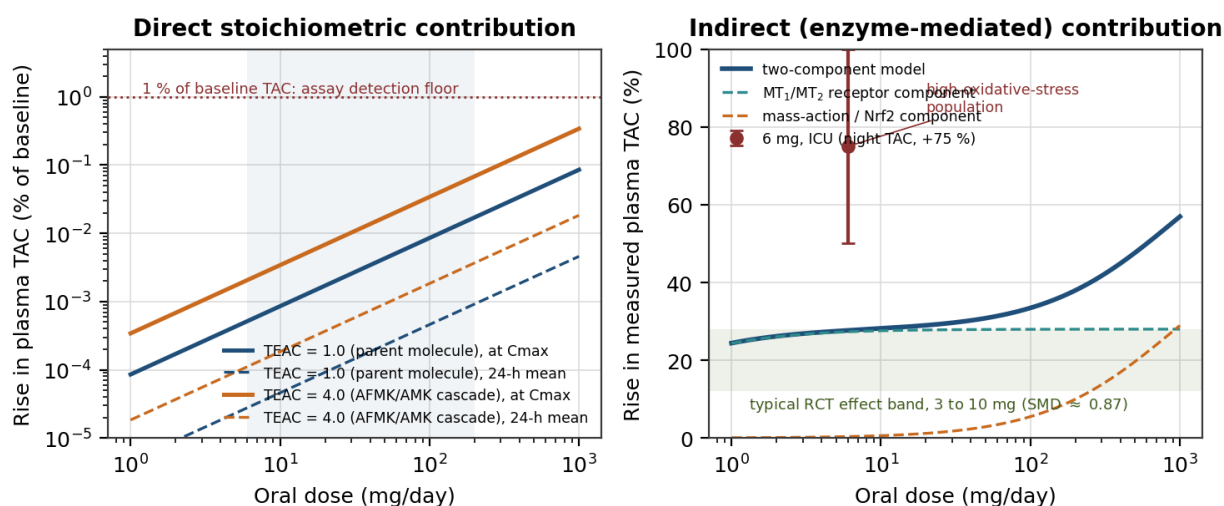


Figure 6: Left: the direct stoichiometric contribution of circulating melatonin to plasma TAC, on log-log axes, for both the parent molecule and the full cascade, at peak and 24-hour mean. It stays four to six orders of magnitude below the 1 % detection floor across the entire dose range. Right: the indirect contribution, modelled as a receptor-mediated component saturating below 1 mg plus a mass-action component with an EC_{50} far outside the achievable range.

redox endpoint would be to chase the mass-action component, and that requires doses where the model itself predicts less than half the maximal effect at 900 mg.

7 What would break this model

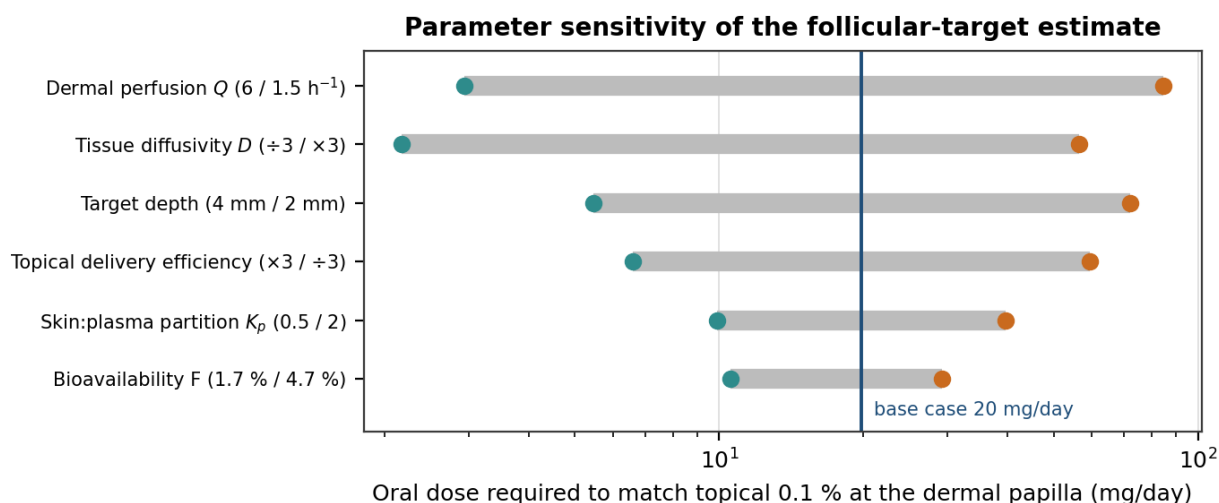


Figure 7: Sensitivity of the follicular-target estimate (base case 20 mg/day to match topical 0.1 % at 3 mm). The transport parameters that set λ dominate; bioavailability and partition coefficient are secondary. The whole plausible range sits between roughly 2 and 90 mg/day, which is to say the qualitative conclusion is robust even though the point estimate is not.

8 Safety notes over this dose range

Melatonin has a wide therapeutic index and doses to 100 mg have been given acutely without serious toxicity, so the constraint above 6 mg is not organ toxicity. It is chronobiological. A 6 mg dose

Assumption	Value used	If wrong
Skin:plasma partition K_p	1.0	Highest-leverage unmeasured parameter. $K_p = 2$ halves the required oral dose and simultaneously halves the topical gradient, so the crossover depth is insensitive to it; the absolute doses are not.
Dermal perfusion Q	3 h^{-1}	Sets λ and therefore the whole geometry. Scalp perfusion is high and follicular perfusion higher still, which shortens λ , deepens the topical penalty, and favours oral further.
Perfusion-limited distribution	assumed	Well supported by melatonin's membrane permeability, but if binding to intracellular targets creates a deep-tissue sink the plateau assumption fails.
Follicular duct bypass	ignored	The largest single omission. If the active site is the upper follicle rather than the bulb, the answer changes from 20 mg to unreachable.
Oral F constant at 2.5 %	fixed	Certainly wrong at high dose; saturable first-pass makes real high-dose exposure larger than modelled, biasing the required doses upward.
Steady state	assumed	Topical is applied once daily and IR oral is over in four hours, so neither is truly steady. The 24-hour mean comparison is the defensible one, which is why the tables use it.

Table 5: Assumption register.

already produces a tenfold supraphysiological excursion and a plasma profile that persists into the biological morning; 50 to 100 mg produces peaks two orders of magnitude above the physiological ceiling with measurable next-day sedation and, at sustained high dose, phase-response effects that can drag the circadian phase rather than anchor it. The dose that best entrains sleep is well below the dose this model says would be needed for a dermatological effect above the follicle, and those two objectives are in direct conflict.

Two interactions are worth flagging in the specific context of a multi-agent regimen. Melatonin is cleared almost entirely by CYP1A2, so any CYP1A2 inhibitor multiplies exposure severalfold, and nicotine cessation removes a CYP1A2 *inducer*, which raises melatonin exposure at fixed dose over the weeks following cessation. That is a moving target, not a fixed one. Separately, melatonin has modest hypotensive and glucose-tolerance effects at high dose. None of this argues against 6 mg. It does argue that if the plan is to test 20 mg or above chronically, that is a conversation to have with your prescriber once rather than a change to layer in silently on top of six existing agents, particularly during a period when one of your clearance pathways is actively shifting.

What the model actually settles

Oral melatonin and topical melatonin are not two routes to the same exposure. They are two different drugs distinguished by which 3 mm of tissue they can reach. For anti-aging, photoprotection, and anything epidermal, oral is not a substitute at any dose and the topical literature's concentrations are simply out of reach. For androgenetic alopecia, if the target is the dermal papilla, oral is not merely adequate but arguably the better delivery route, and 20 mg SR reproduces the strongest topical formulation ever tested in that indication while 6 mg already exceeds the weakest one. And for total antioxidant capacity, the dose question is a non-question: the receptor is saturated at 1 mg and everything above that is a rounding error on a plateau.

Sources: Andersen LPH et al., *Pharmacokinetics of oral and intravenous melatonin in healthy volunteers*, *BMC Pharmacol Toxicol* 2016;17:8. Harpsøe NG et al., *Clinical pharmacokinetics of melatonin: a systematic review*, *Eur J Clin Pharmacol* 2015;71:901. Fischer TW et al., *Percutaneous penetration of topically applied melatonin in a cream and an alcoholic solution*, *Skin Pharmacol Physiol* 2004;17:190. Fischer TW et al., *Melatonin increases anagen hair rate in women with androgenetic alopecia or diffuse alopecia*, *Br J Dermatol* 2004;150:341. Fischer TW et al., *Topical melatonin for treatment of androgenetic alopecia*, *Int J Trichology* 2012;4:236. Scheuer C et al., *topical melatonin safety and solar erythema studies*, 2016. Dreher F et al., *Effect of topical antioxidants on UV-induced erythema*, *Br J Dermatol* 1998;139:332. Bangha E et al., *Suppression of UV-induced erythema by topical melatonin*, *Dermatology* 1997. Mistràletti G et al., *Melatonin blood values and total antioxidant capacity in critically ill patients*, *Crit Care* 2014 / PMC4069407. Mohammadi S et al., *Effect of melatonin supplementation on cardiometabolic risk factors, oxidative stress and hormonal profile in PCOS*, *J Ovarian Res* 2024;17:145. Tan DX et al., *Mechanistic and comparative studies of melatonin and classic antioxidants in terms of their interactions with the ABTS cation radical*, *J Pineal Res* 2003;34:249. Yu H et al., *High membrane permeability for melatonin*, *J Gen Physiol* 2016;147:63. Karasek M / Reiter RJ reviews for MT₁/MT₂ affinity and Nrf2 coupling. *Topical trial compilation from Int J Mol Sci* 2024;25:5167, *Clinical Studies Using Topical Melatonin*.